

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE – NEW**

3410U20-1



S18-3410U20-1

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

FOUNDATION TIER

THURSDAY, 17 MAY 2018 – MORNING

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper
 you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen
 or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional
 page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be
 assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common
 ions on the inside of the back cover.

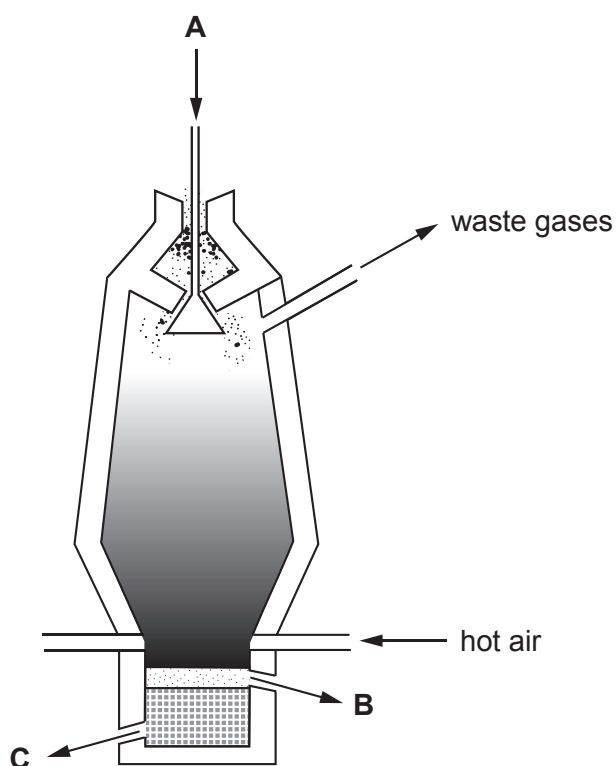
For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	12	
2.	10	
3.	10	
4.	9	
5.	6	
6.	13	
7.	8	
8.	12	
Total	80	

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01

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Answer **all** questions.

1. (a) The diagram shows where substances enter and leave the blast furnace in the extraction of iron from iron ore.



coke

slag

iron

limestone

iron ore

- (i) Use the substances in the box to complete the following sentences. [3]

The **three** raw materials which enter the furnace at **A** are ,
..... and

Product **B** is and product **C** is



(ii) One reaction that takes place in the furnace is



Underline the element which is removed from the iron(III) oxide during the reaction. [1]

iron

oxygen

carbon

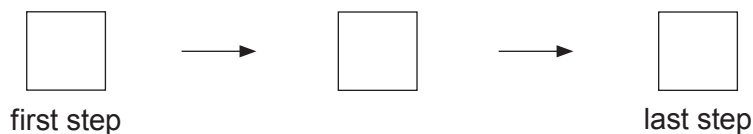
(b) Statements **D**, **E** and **F** show the three steps needed to prepare a sample of copper(II) chloride in the laboratory. The steps are **not** in the correct order.

D filter to remove excess copper(II) oxide

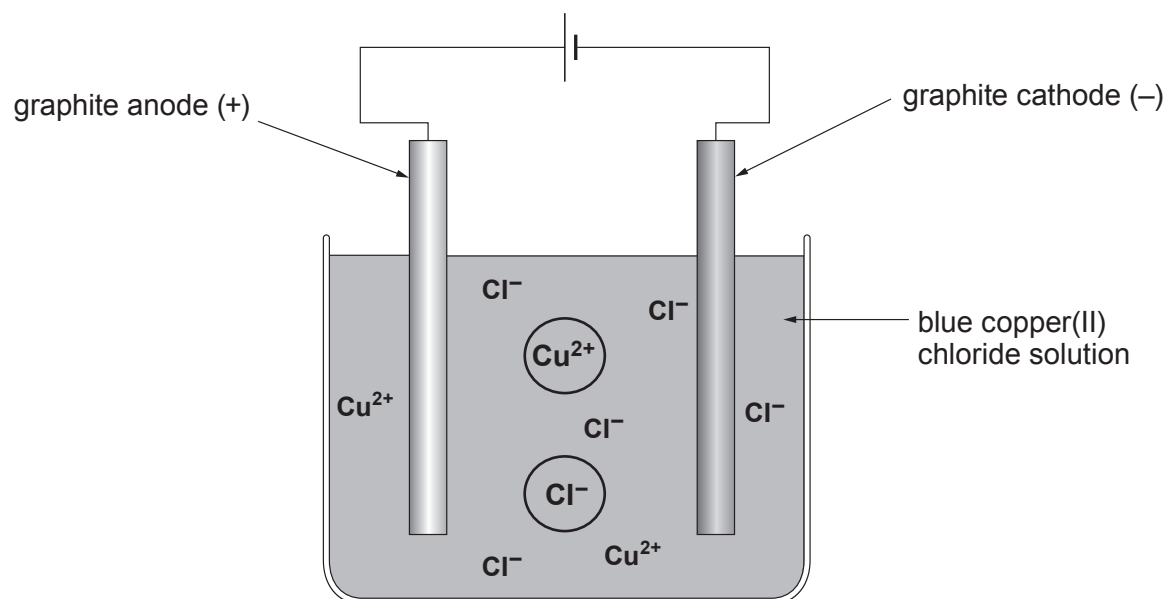
E leave the copper(II) chloride solution to evaporate at room temperature

F add excess copper(II) oxide to dilute hydrochloric acid

Complete the flow chart by putting the **letters** in the correct order. [2]



(c) The diagram shows the apparatus used to obtain copper from copper(II) chloride solution.



electrolyte	electrodes	electricity	electrolysis
-------------	------------	-------------	--------------

- (i) Choose words from the box to complete the following sentences. [2]

The breaking down of copper(II) chloride solution using an electric current is called

.....

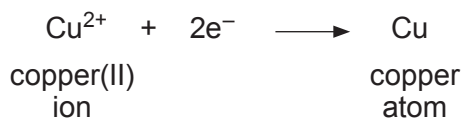
Electric current enters and leaves the copper(II) chloride solution by the

.....

- (ii) **By drawing arrows on the diagram**, show the direction in which the **circled** ions move during the process. [1]



(iii) The reaction occurring at the cathode is:



I. Put a tick (✓) in the box next to what e^{-} represents in the equation. [1]

electricity

☐

electron

☐

ion

☐

atom

☐

II. Copper(II) ions, Cu^{2+} , are removed from the solution during the process.

Put a tick (✓) next to the statement which describes what, if anything, happens to the colour of the copper(II) chloride solution. [1]

the blue solution turns darker

☐

the blue solution turns paler

☐

the blue solution turns yellow

☐

nothing happens to the colour of the blue solution

☐

(iv) Put a tick (✓) next to the gas which is formed at the anode. [1]

oxygen

☐

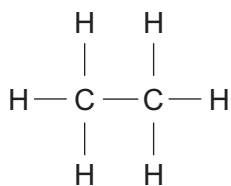
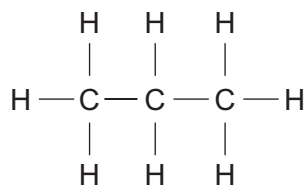
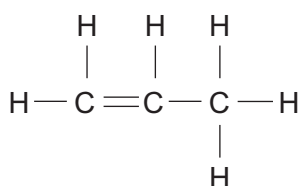
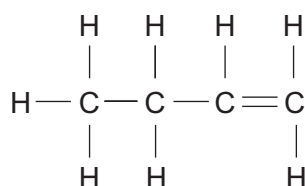
hydrogen

☐

chlorine

☐


2. (a) The structural formulae of four carbon compounds are shown below.

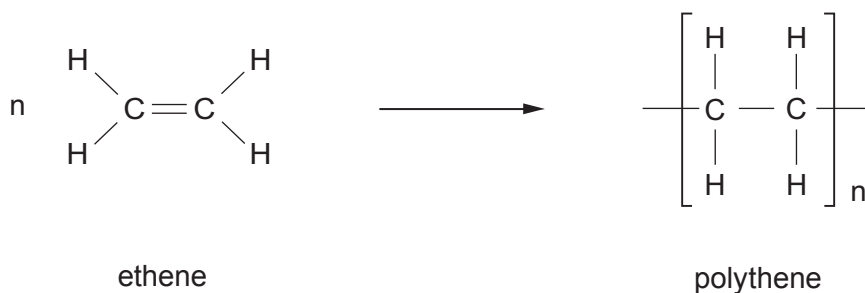
**A****B****C****D**

Complete the table by choosing the **letter A, B, C or D** which represents the structural formula of the named compounds. [2]

Name	Molecular formula	Structural formula
ethane	C_2H_6	
propene	C_3H_6	

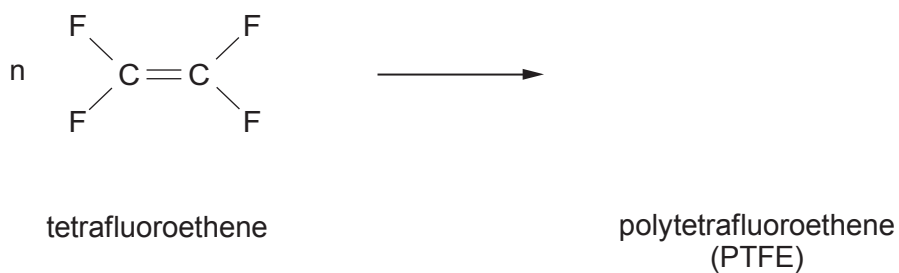


(b) The equation below shows the formation of polythene from ethene.

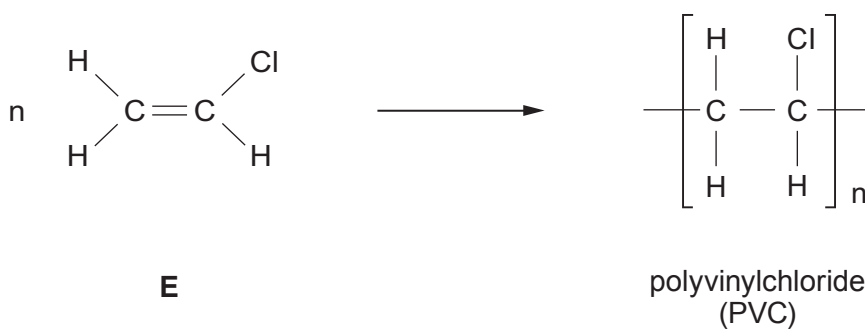


Use this information to complete parts (i) and (ii).

(i) Complete the equation for the formation of polytetrafluoroethene (PTFE) from tetrafluoroethene. [1]



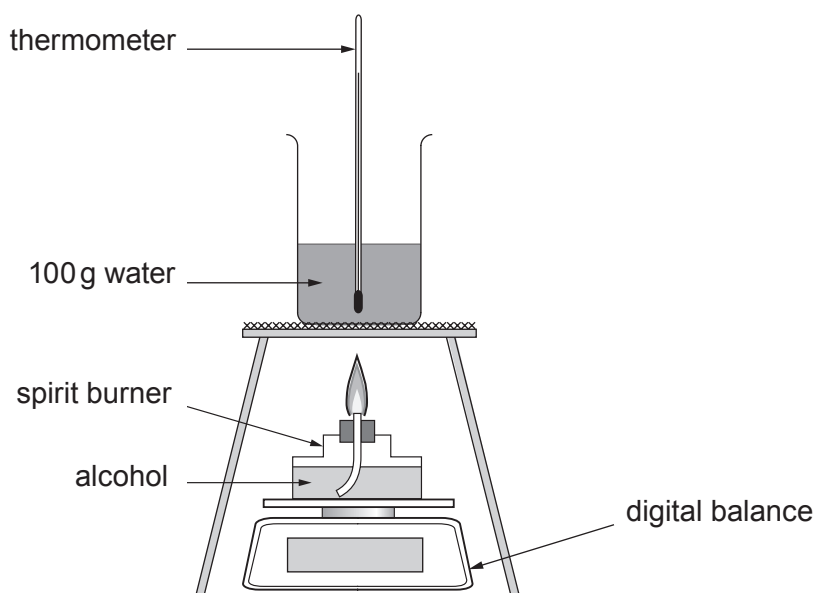
(ii) Name monomer **E**. [1]



E



- (c) A student carried out an investigation to find which alcohol gives out the most energy when burned.



100 g of water was heated by burning 1 g of each alcohol.

Alcohol	methanol	ethanol	propanol
Formula	CH ₃ OH	C ₂ H ₅ OH	C ₃ H ₇ OH
Relative molecular mass (M_r)	?	46	60
Temperature of water before heating (°C)	20	21	20
Temperature of water after heating (°C)	45	58	60
Energy given out (J)	?	15 700	17 200

- (i) Calculate the relative molecular mass (M_r) of methanol, CH₃OH.

[2]

$$A_r(\text{H}) = 1 \quad A_r(\text{C}) = 12 \quad A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$



- (ii) The energy given out can be calculated using the formula:

$$\text{energy given out} = \text{mass of water} \times 4.2 \times \text{temperature change}$$

Use the data given to calculate the energy given out when burning methanol. [2]

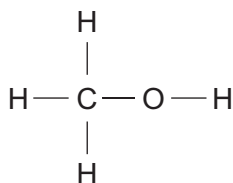
Energy given out = J

- (iii) Give the **letter** of the correct conclusion for the student's investigation. [1]

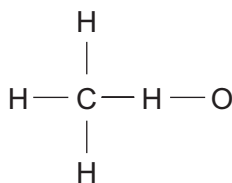
- A** all alcohols burn giving out the same amount of energy
- B** the greater the number of carbon atoms in the alcohol molecule the less energy is given out
- C** the greater the number of carbon atoms in the alcohol molecule the more energy is given out
- D** as the number of carbon atoms doubles the amount of energy given out doubles

Letter

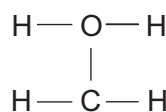
- (iv) Give the **letter** of the structural formula of methanol, CH₃OH. [1]



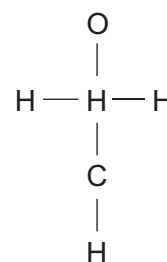
A



B



C



D

Letter



3. (a) Crude oil can be separated into simpler mixtures called fractions. These fractions contain hydrocarbon compounds called alkanes. **Table 1** shows information about some of the fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ -C ₄
petrol	30-75	C ₅ -C ₁₀
naphtha	70-170	C ₈ -C ₁₂
kerosene	170-250	C ₁₀ -C ₁₄
diesel oil	250-340	C ₁₄ -C ₂₄
lubricating oil	340-500	C ₂₁ -C ₃₀
fuel oil	490-580	C ₂₅ -C ₃₅
residue	> 580	>C ₃₅

Table 1

Use **only** the information in **Table 1** to answer parts (i)-(iii).

- (i) Hexane has a boiling point of 68 °C. Give the name of the fraction which contains hexane. [1]

.....

- (ii) One alkane is found in kerosene and in diesel. Give the number of carbon atoms in this alkane. [1]

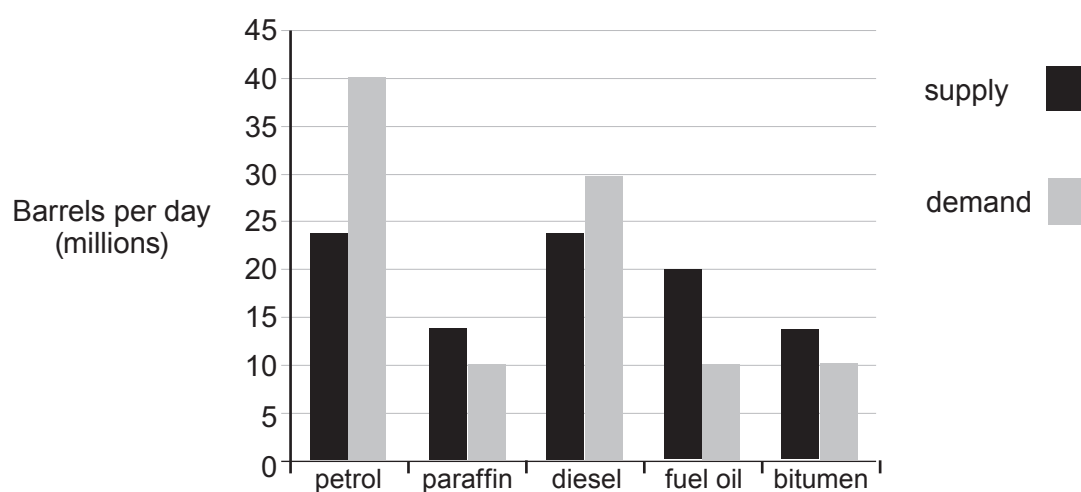
.....

- (iii) Give the number of carbon atoms in the alkane which has the **lowest** boiling point. [1]

.....



(b) The bar chart shows the **supply** and **demand** for some crude oil products.



Name the fractions for which the demand is greater than the supply. Suggest a reason why these fractions are in high demand. [2]

Fractions and

Reason

.....



- (c) Plastic carrier bags are made from polythene. Each plastic carrier bag can take 500-1000 years to decompose and may never break down in landfill. Paper bags are not necessarily an environmentally friendly alternative. Manufacturing paper bags wastes a lot of natural resources. Even starch-based biodegradable bags use natural resources during their manufacture.

Supermarkets give customers a choice of buying single-use or re-usable polythene carrier bags.

Table 2 shows the number of both types of plastic bag sold in UK supermarkets from 2011 to 2013.

Year	2011	2012	2013
	Number of bags (millions)		
Single-use bags	7977	8079	8455
Re-usable bags	415	408	445

Table 2

- (i) State **one** environmental problem related to the disposal of **all** types of carrier bag. [1]

.....

- (ii) Calculate the percentage of plastic bags sold in 2013 that were single-use bags. [2]

Percentage = %



- (iii) Although more plastic carrier bags were sold in 2013 than 2012, the total **mass** of those bags changed from 70 400 tonnes to 67 300 tonnes.

Put a tick (✓) in **two** boxes next to statements which could explain the reason for the change in mass. [2]

the bags were made the same thickness but from a less dense plastic

☐

customers re-used their plastic bags more often

☐

the bags were made from the same plastic but were thicker

☐

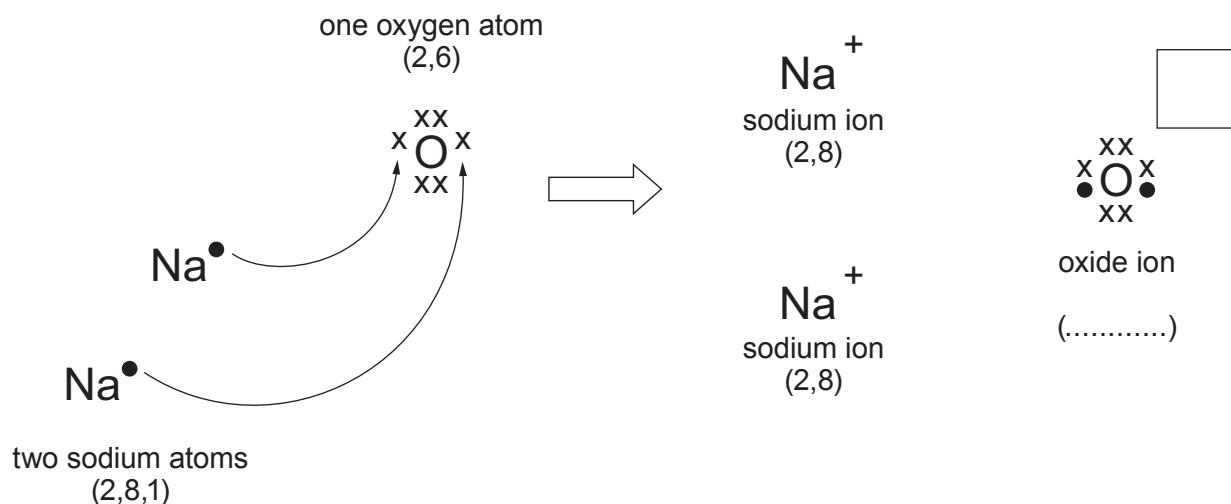
the bags were made from the same plastic but were thinner

☐

the bags were made the same thickness but from a more dense plastic

☐


4. (a) The diagram shows the electronic changes that occur when sodium reacts with oxygen to form sodium oxide. The ● and x symbols are outer shell electrons.



- (i) **Complete the diagram** by putting in the electronic structure **and** the charge of the oxide ion. [2]
- (ii) Complete and balance the **symbol** equation for the reaction between sodium and oxygen. [2]



- (b) The table shows the electronic structure of the elements present in water and hydrogen chloride.

Element	Electronic structure
hydrogen	1
oxygen	2,6
chlorine	2,8,7

The diagram shows the bonding in a water molecule.



Give the **letter** of the diagram which shows the bonding in a hydrogen chloride molecule. [1]



A



B



C

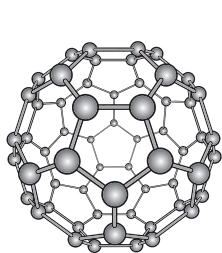


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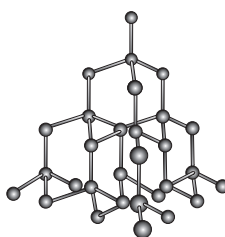
Letter



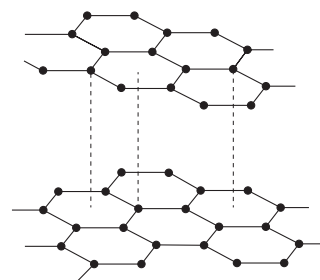
(c) Fullerene, diamond and graphite are different forms of carbon.



fullerene



diamond



graphite

(i) Underline the structure of these three forms of carbon.

[1]

simple molecular

giant covalent

giant ionic

(ii) The box shows some properties of different forms of carbon.

transparent

insoluble in water

conducts electricity

hollow

soft

does not conduct electricity

hard

Complete the table by choosing the property which makes the different forms of carbon suitable for the uses shown. [3]

Form of carbon	Use	Property
diamond	drill bits	
graphite	pencils	
fullerene	drug delivery	



Examiner
only

5. Every year thousands of acres of moorland are destroyed by fires in Wales. Firefighters use several methods to put out this type of fire.

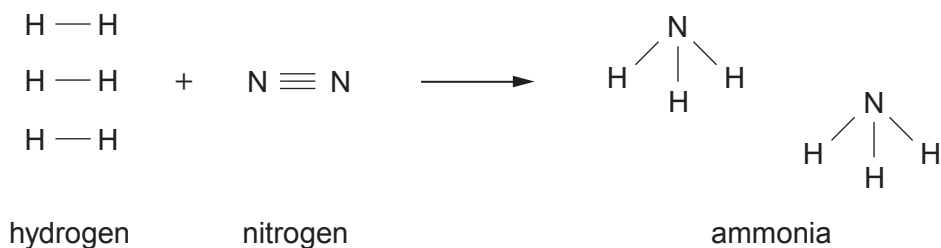


State and explain, in terms of the fire triangle, **three** methods that are used to put out moorland fires. Each method must refer to a **different** part of the fire triangle. [6 QER]



6. Ammonia is manufactured from hydrogen and nitrogen in the Haber Process.

- (a) The equation shows the bonds which are broken and the bonds which are formed during the manufacture of ammonia.



The relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
$\text{N} \equiv \text{N}$	945
$\text{H} - \text{H}$	436
$\text{N} - \text{H}$	391

- (i) Calculate the **total** energy needed to break **all** the bonds in the hydrogen molecules and the nitrogen molecule. [2]

Energy = kJ

- (ii) Calculate the **total** energy released when **all** the bonds in the ammonia molecules are formed. [2]

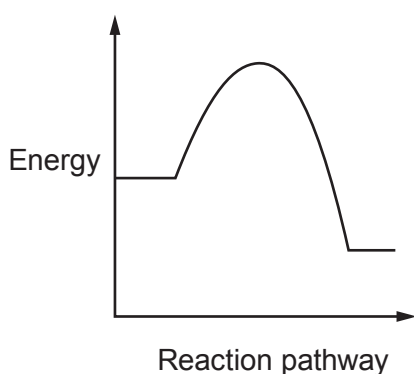
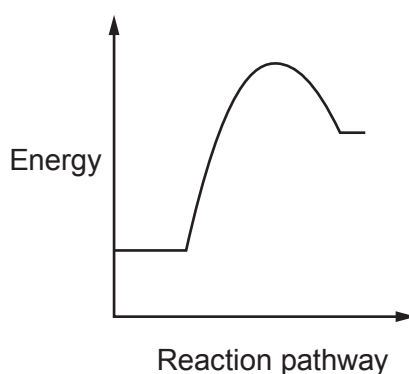
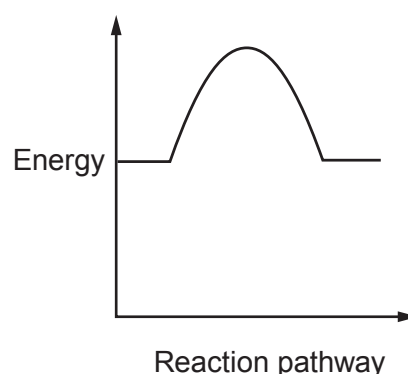
Energy = kJ

- (iii) Use your answers to parts (i) and (ii) to calculate the overall energy change. [1]

Energy change = kJ



- (b) The manufacture of ammonia is an exothermic reaction. Put a tick (✓) in the box next to the energy profile diagram that shows an exothermic reaction. [1]


☐

☐

☐

- (c) The table shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)				
	100	200	300	400	500
	Yield of ammonia (%)				
100	96.7	81.7	52.5	25.2	10.6
200	98.4	89.0	66.7	40.0	18.3
400	99.4	94.6	79.7	55.4	31.9

Use the information in the table to answer parts (i) and (ii).

- (i) State what happens to the yield of ammonia as the temperature increases. [1]

- (ii) One manufacturer carries out the Haber Process at 200 atm and 450 °C.

Underline the approximate percentage yield of ammonia formed under these conditions. [1]

10 %

30 %

40 %

58 %



(d) Ammonia is used in the manufacture of nitrogenous fertilisers. One example of a nitrogenous fertiliser is ammonium nitrate. Ammonium nitrate is formed by the reaction between an acid and ammonia.

(i) Complete the **word** equation by naming the acid used in this reaction. [1]

ammonia + $\longrightarrow$ ammonium nitrate

(ii) When ammonium sulfate solution is warmed with sodium hydroxide solution a pungent gas is formed. Damp red litmus paper is used to test this gas.

I. Describe the change in colour of the litmus paper during the test. [1]

.....

II. State the property of this gas which causes the colour change. [1]

.....

III. Name the gas formed. [1]

.....

(iii) Nitrogenous fertilisers pollute streams and rivers. State how nitrogenous fertilisers get into these waterways. [1]

.....

.....



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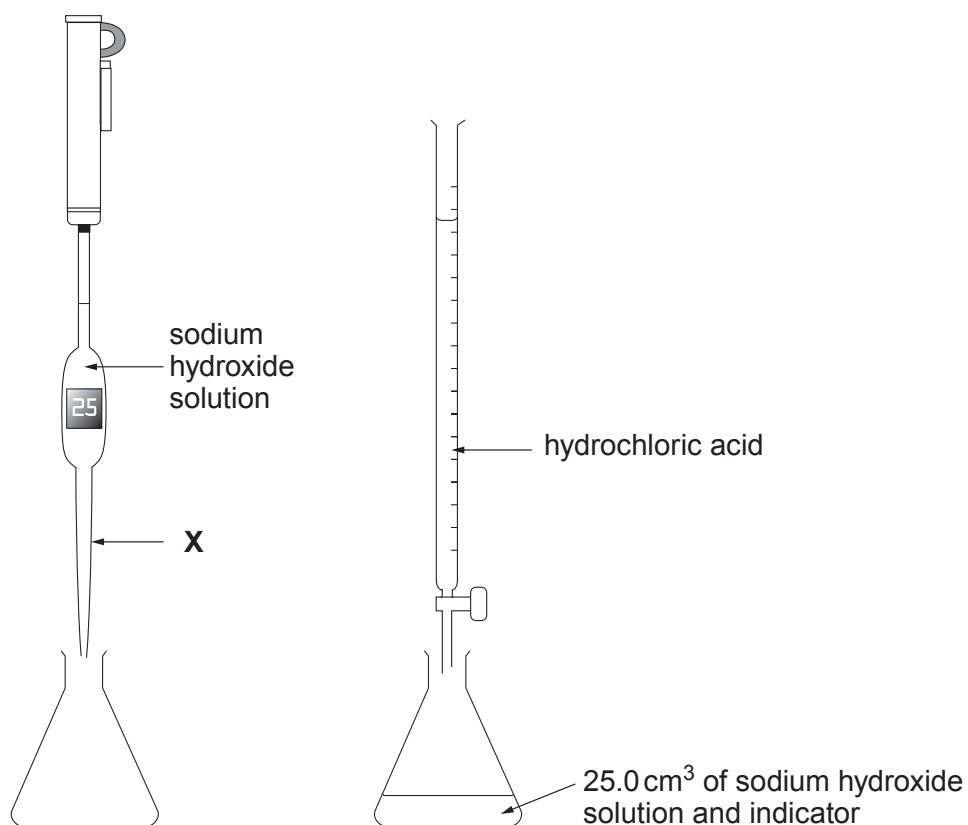
7. (a) Hydrochloric acid, HCl , reacts with sodium hydroxide solution to form sodium chloride and water only.

Write a balanced **symbol** equation for this reaction.

[2]

- (b) A group of students was asked to find the volume of hydrochloric acid solution needed to neutralise 25.0 cm^3 of sodium hydroxide solution. They decided to titrate sodium hydroxide with hydrochloric acid.

Apparatus



Results

Titre	1	2	3
Volume of hydrochloric acid needed (cm^3)	18.2	17.8	18.0



- (i) Name the piece of apparatus **X**. [1]

.....

- (ii) Explain the purpose of the indicator. [1]

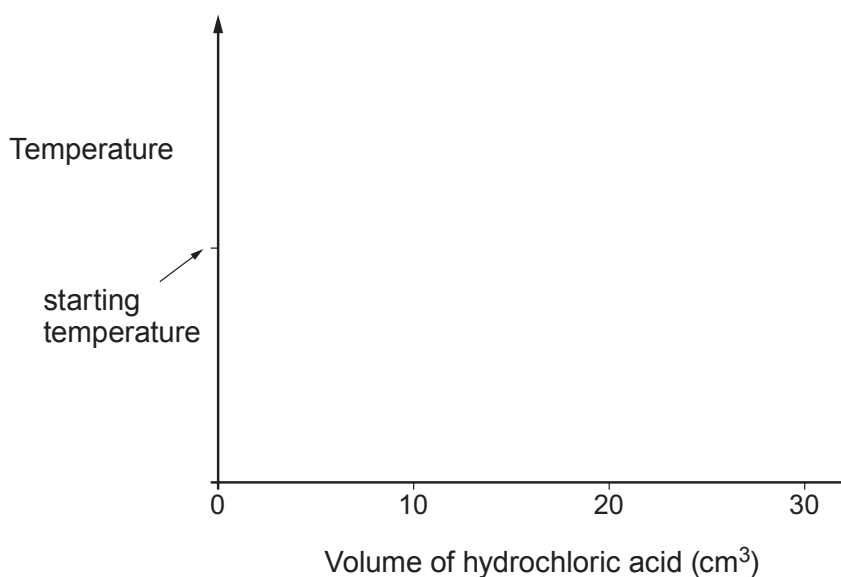
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- (iii) Calculate the mean volume of hydrochloric acid needed to neutralise 25.0 cm^3 of the sodium hydroxide solution. [1]

Mean volume = cm^3

- (iv) The change in temperature during the reaction can be monitored using a temperature sensor.

Sketch a graph on the axes below to show how the temperature changes as more and more acid up to a total of 30 cm^3 is added. [2]



- (v) The experiment was repeated using hydrochloric acid of **half** the original concentration.

State the volume of hydrochloric acid that would be needed to change the indicator colour. [1]

Volume = cm^3



8. (a) When a mixture of iron(III) oxide and aluminium powder (Thermit mixture) is heated, there is a violent reaction. The reaction is carried out in a tube surrounded by a mound of sand because the temperature reaches 2500°C . A bead of iron is recovered from the sand. The picture below shows the reaction taking place in a darkened room.



- (i) Give the reason why the iron formed in the reaction is molten. [1]

.....
.....

- (ii) Complete and balance the **symbol** equation for this reaction. [2]



- (iii) State which of the substances is oxidised. Give the reason for your choice. [1]

.....
.....

- (iv) When a mixture of magnesium oxide and aluminium powder is heated, there is no reaction.

List iron, magnesium and aluminium in order of reactivity. [1]

Most reactive

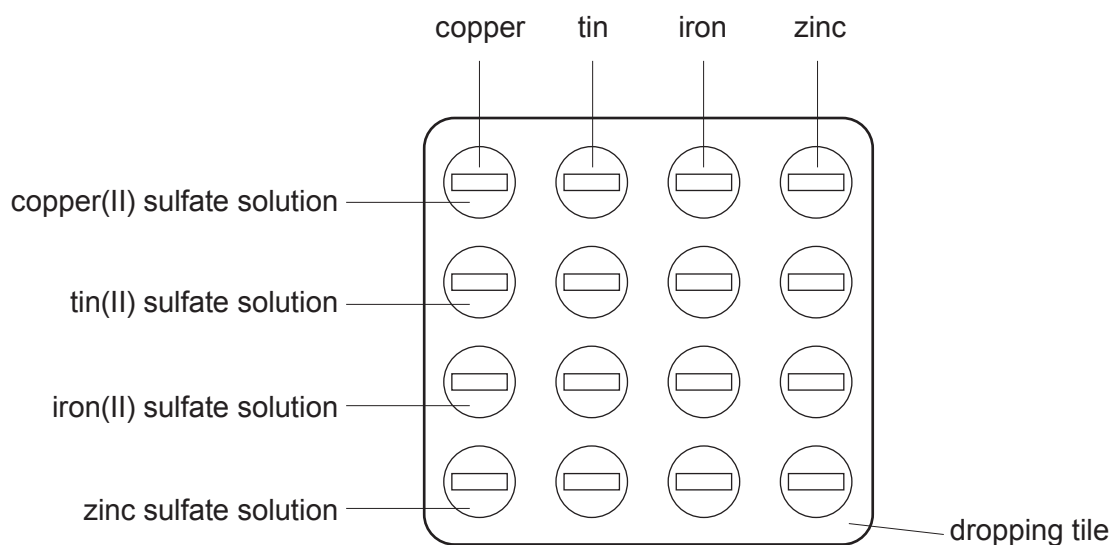
.....

Least reactive



- (b) Some metals are more reactive than others. A more reactive metal displaces a less reactive metal from its compounds.

A student was given tin, iron, copper and zinc and solutions of the metal sulfates. Using a dropping pipette, she put a little of one of the sulfate solutions in four of the depressions of the dropping tile. She did this for each solution in turn. She then put a piece of metal foil in each of the solutions, as shown below.



- (i) Put a tick (✓) next to the question which **best** describes the investigation the student is carrying out. [1]

Which displacement is the most exothermic?

☐

Which metal can displace copper from solution?

☐

What is meant by the reactivity series?

☐

What are the positions of the four metals in the reactivity series?

☐


- (ii) The student recorded the results by putting a tick (✓) next to a mixture which showed signs of a reaction and a cross (X) next to a mixture which showed no signs of a reaction.

The student concluded that:

tin displaces copper
iron displaces tin
iron displaces copper
zinc displaces iron

Give the **letter** of the tile below which shows the results she recorded.

[1]

Letter

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

A

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

B

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

C

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

D

- (iii) Another student said that not all of the tests were necessary. Give **one** example of a test not needed. Explain your choice.

[2]

Example

Explanation

.....



- (c) Copper displaces silver from a solution of silver nitrate, AgNO_3 , to form copper(II) nitrate solution.

(i) Describe **one** change the student would **see** during this displacement reaction. [1]

.....

.....

(ii) Write a balanced **symbol** equation for this reaction. [2]

.....

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FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2																									
7 Li Lithium 3		9 Be Beryllium 4		11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10																					
23 Na Sodium 11		24 Mg Magnesium 12		27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18																					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		127 I Iodine 53		131 Xe Xenon 54			
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number